

Annotation

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MSAI, fall 2024

Outline

- Annotation pipeline
 - overlap
 - honeypots
- People management
 - inhouse
 - crowd
- Instructions
- Solutions for fields
 - CV
 - NLP

What to annotate?

- Tabular data
 - classification, regression
- Texts
 - word type
- Images
 - segmentation, detection, skeletons
- Videos
 - object tracks
- 3D data (point clouds, meshes)
 - 3d detection, segmentations
- Sounds
 - subtitles, genre of music

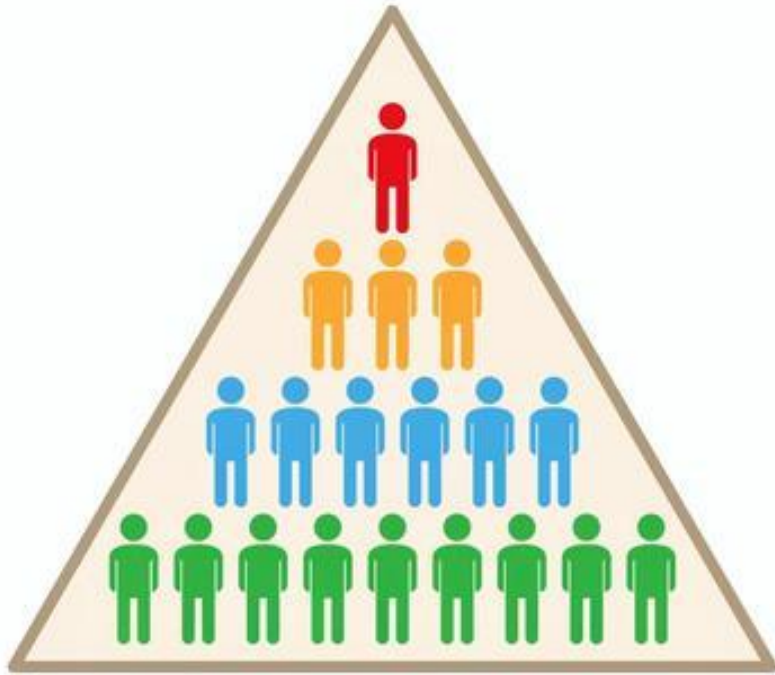


What to annotate

- Таблицы
- Временные ряды
- Текст
- Картинки
 - Классификация
 - OCR
- Видео
- 3D данные
 - Лидары
 - Mesh
- Звук
 - Текст
 - Классификация отрезков
- Мультимодальные данные



Task management



1. Owner
 - a. responsible for the whole pipeline
2. Experts
 - a. data engineers
 - b. ML engineers and analytics
 - c. model users (non technicians)
3. Leaders
 - a. Supervise assessors
 - b. label data
4. Assessors
 - a. label data by instruction



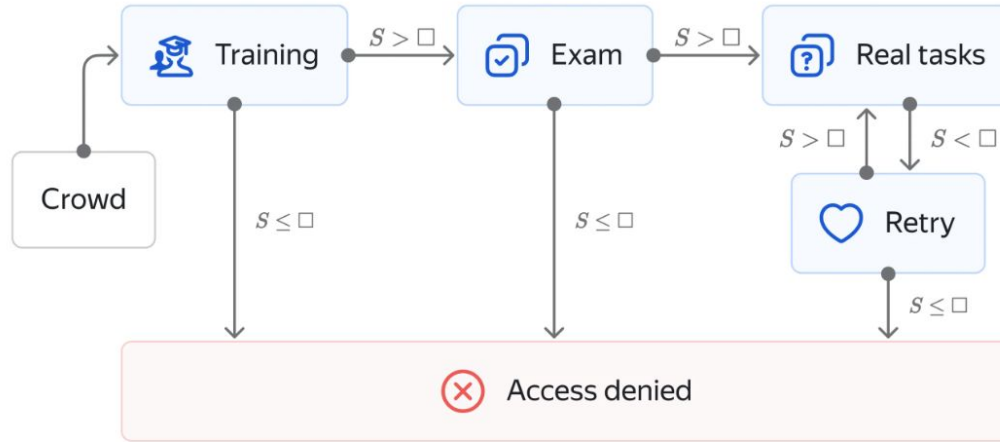
Who are the assessors?

People from regions with low motivation.

Usually retired or students



Assessor lifecycle



1. Learning
2. Qualification
3. Tasks completions
4. Requalification

To rank assessor we use skill level measuring our trust

Requalification can happen because of absence

Honeypots aka golden set

In computer terminology, a honeypot is a computer security mechanism set to detect, deflect, or, in some manner, counteract attempts at unauthorized use of information systems.

[Common knowledge site](#)

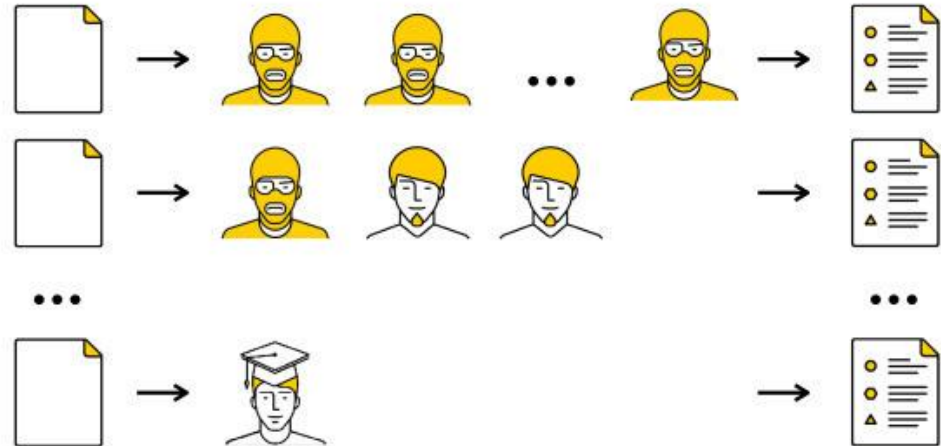


Overlap

For best result we need to annotate the same sample several times.

Usually it's 3 people per sample with Majority Vote aggregation.

Also there are advanced aggregation techniques which requires setup for each single case

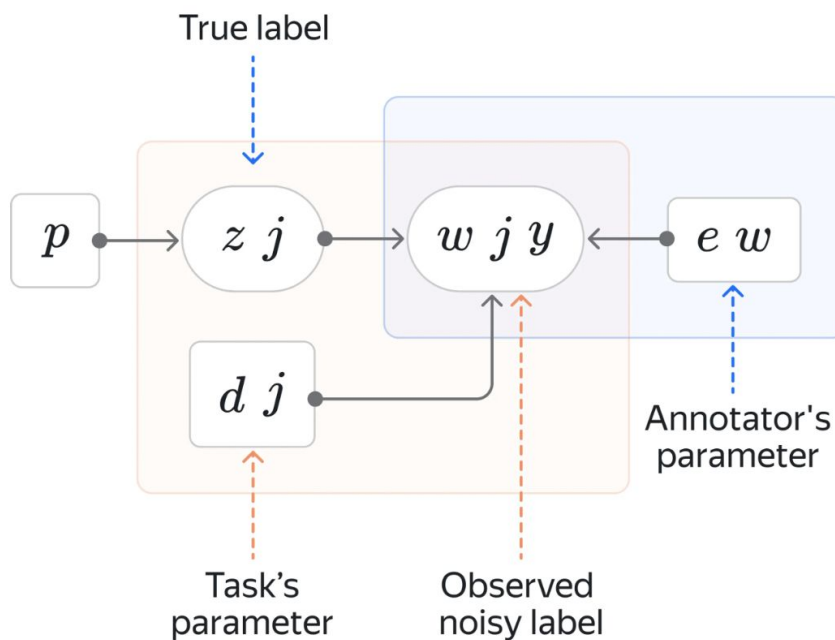


Aggregation

For best result we need to annotate the same sample several times.

Usually it's 3 people per sample for aggregation.

Also there are advanced aggregation techniques which requires setup for each single case



Bidding

Quality of labeling barely depend on money you pay



People management

	Inhouse	Crowd
Scalability	low	high
Quality	high	low
Cost	moderate	low

- For crowd per task cost does not really matter after some threshold
- Experts should be employed full time, Assessors paid per label

How much to pay?

1. Budget we have and required amount of labels
2. Look for existing tasks
3. Estimate it yourself based on average salary and time consumption
 - a. first as honest person
 - b. then as frauder



Relevance levels

2 documents

- vital
- rel+
- rel
- rel-
- not rel

[Google's guide for query annotation](#)



Project management

Project

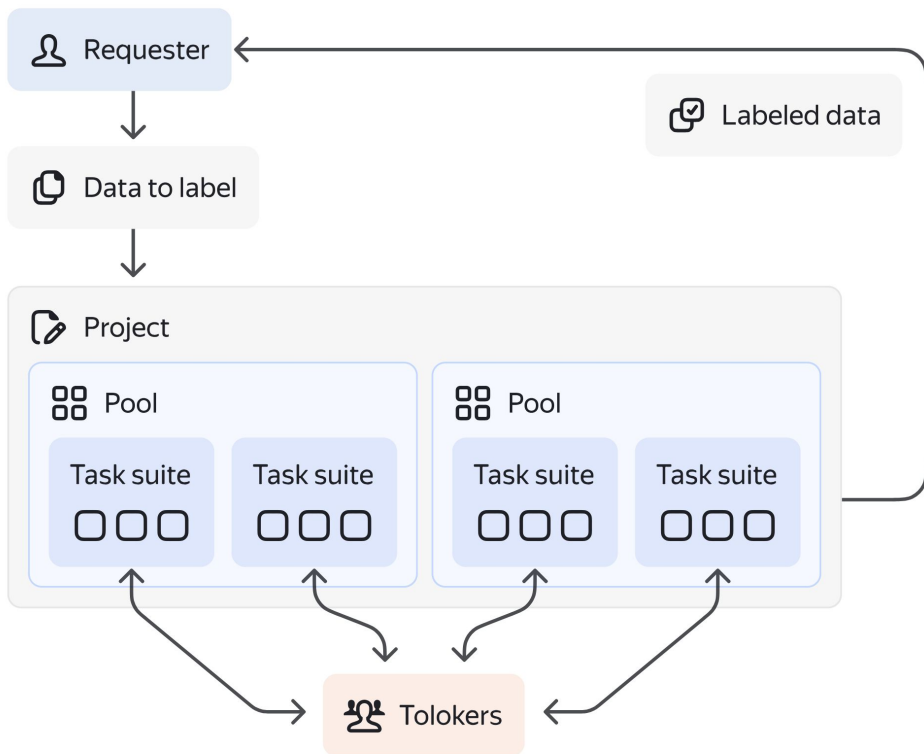
Main description
Interface
Task instructions
Input and output
data specifications

Pool

Price
Toloker selection
Overlap
Quality control

Task

Task data
text, image URLs, and more
Correct responses
Tips for training



Tools

Crowd:

- Toloka
- MTurk Amazon
- crowds of them...

Self hosted:

- standalone ([VIA](#))
- server based ([CVAT](#))
- <https://github.com/HumanSignal/label-studio>
- lots of them

Overviews: [one](#), [two](#)



Toloka

<https://www.youtube.com/watch?v=gkjOH5SYkuU&list=PLJOzdkh8T5kq9eSxQhUJ63HH68oPj532O>

tasks available in toloka <https://toloka.ai/docs/tutorials/usecases/>

structure <https://toloka.ai/docs/guide/overview/>
<https://toloka.ai/technologies/>



Other topics

доклад о разметчиках https://youtu.be/gHi6lFUs6_Q



Thanks for attention!

Questions?

